Auto collect:

- User behaviour
- Device Fingerprint
- Network info
- Location /IP
- API calls
- UI integrations
- Risk signals

Sub-2ms Request Overhead (90% request)

No GPU ONNX on standard CPU

1M+ events/sec kafka+flink

All Vendors AWS, Azure, GCP , On-Prem, SaaS

3-Second Plain-English Incident Report

Global Enforcement 1 Redis,Multiple Gateway Plugins

LLM <1% Under 1,000 calls per million sessions

Cross-Platform Works for web ,mobile, legacy, AI agents

HTTPS Request

USERS & CLIENTS

- **web browser** (JS JDK)
- **Mobile App** (IOS/Android SDK)
- **Desktop App** (Windows/MAC)
- **AI Agents** (API clients)
- **Third-Party Apps** (SaaS/Partners)

L1: Browser SDK & Signal Collection Layer (Oms,Async)

ZeroWatch SDK

- **Fetch/XHR Interceptor** (auto-signs requests)
- **FingerprintsJS** (device fingerprint)
- **ECDSA Key Pair** (stored in IndexedDB)
- **MaxMind Geolocation** (IP,ASN,VPN)
- **Behavioural Signals** (mouse, keystrokes, navigation)
- **Custom Events** (API, calls, errors, logs)

Send Events (Async) user + device + request+ Behavioural Signals

L2: Stream Processing & Risk Scoring Layer (Apache flink + Redis

Kafka (Event Ingestion)

Apache Flink (Real-time Processing)

- Feature Extraction
- Isolation Forest
- Z-score Calculation
- Behavioural Analysis
- ONNX Model Inference (sub-5ms-CPU
- Data Enrichment

Redis Cluster (Risk Score Cache)

- Risk Score (0-100)
- TTL(e.g., 5mins)
- HA (Cluster)
- Global Replication(sub-50ms)
- Device Binding Info
- Session State

Day-1 LLM Synthetic Baseline (at account creation)

Behavioural Model (Isolation Forest+ONNX)

Continuous Learning (Update baseline periodically)

Risk Trajectory (Predict next hour)

L3 : LLM Analysis Layer (Async,<1% Requests)

AI query (Anthropic)

Cold-state Analysis

Generate user baseline at account creation

Incident Narration

Convert z-scores to plain-English explanation in ~3 seconds

Intent Classification

Analyse API call sequence and classify intent (malicious/legitimate)

L4: Communication & Alerting Layer (Twilio Integration)

twilio

- **SMS Alerts**
- **Voice Calls**
- **WhatsApp Notifications**
- **Email**
- **Programmable Messaging**

Use Cases

- Security alert to user
- MFA/Step-up Authentication
- Account compromise notification
- Incident report sharing (link)

L5 :cement Layer (API Gateway + Service Mesh)

API Gateway Plugins (AWS/Azure/GCP/ON-prem)

- Read risk score from Redis
- Verify ECDSA Signature
- Enforce policies (Allow/Block/Challenge)
- Add X-ZeroWatch-Score header
- Step-up authentication (MFA)

Service Mesh(Istio)

- Envoyfilter (inject score)
- Enforce for all internal services
- works for legacy apps, microservice SaaS and AI agents.
- NO app code changes required
- mltmLS between services

Project Apllications

- **Legacy Applications** (Monolithhs)
- **Microservices** (Kubernetes)
- **SaaS Applications** (Salesforce, Workday, etc.)
- **AI Agents** (LLM Tools)
- **Databases** (Internal/External)

L6 :Data & Storage Layer

- **PostgreSQL** (Session, user profiles, incident logs)
- **Redis** (Real-time risk score, agent state)
- **Object Storage** (Logs, raw events, model artifacts)
- **Vector Store** (Embeddings, semantic search)
- **Elasticsearch** (Logs & Analytic)
- **Data Lake** (Historical analysis, training data

L7 : Monitoring, Observability & SIEM Integration

- **Prometheus** (Metrics)
- **Open Search** (Search & Analysis)
- **SEM/SOAR** (Sentinel, splunk, QRadar,etc.)
- **Alert Manager** (Rules & Notifications)

L8 : AI Models & Tools Layer

- **LLM Providers**
- **ONNX Runtime**
- **Tools & Integrations**
- **Prompts Templates**
- **Model Router**

L9 : Infrastructure & Orchestration Layer (Kubernetes)

Container Orchestration

Kubernetes (HPA, Auto-scaling, Self-healing)

CI/CD Pipeline

GitHub Actions / GitLab CI (Build , Test, Deploy

Infrastructure (Multi-cloud)

- **AWS**
- **Azure**
- **GCP**
- **On-Prem**

Security & Networking

- **Istio** (Service Mesh)
- **mTLs** (Service-to-Service
- **Network Policies** (Isolation)
- **Secrets Management** (Vault/KMS)

Global Deployment

Multi Region (Low latency, HA)